

TURNU-MAGURELE, Romania

WMO: 154900

Lat: 43.7603N

Lon: 24.8783E

Elev: 104

StdP: 14.64

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.7	14.4	5.4	7.3	9.7	10.8	9.6	15.5	22.3	33.8	18.2	34.1	1.7	320	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	94.5	72.7	91.6	71.4	89.0	70.2	75.0	90.6	73.4	87.7	72.0	85.2	3.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.4	112.5	79.8	69.1	107.3	78.9	67.8	102.7	77.7	38.5	90.9	37.0	87.6	35.8	85.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	13.4	10.6	2.4	100.1	6.2	3.3	-2.1	102.5	-5.7	104.5	-9.1	106.4	-13.6	108.8
			2.1	78.4	6.1	2.4	-2.3	80.1	-5.9	81.5	-9.3	82.9	-13.7	84.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.8	30.6	38.1	45.8	55.4	64.6	72.4	75.6	75.4	67.1	54.7	43.4	34.1	(6)
	DBStd	17.17	8.72	9.02	7.42	6.98	5.91	5.09	4.59	5.21	7.07	7.51	8.00	8.47	(7)
	HDD50	1903	601	337	169	31	0	0	0	0	0	40	230	495	(8)
	HDD65	4796	1066	752	596	295	80	6	0	4	63	326	649	959	(9)
	CDD50	3667	0	5	38	192	452	671	794	787	512	185	30	1	(10)
	CDD65	1088	0	0	0	6	67	228	330	326	125	6	0	0	(11)
	CDH74	11255	0	0	10	135	727	2270	3346	3409	1242	113	3	0	(12)
	CDH80	4733	0	0	1	25	173	912	1516	1613	473	20	0	0	(13)
(14) Wind	WSAvg	3.5	3.4	3.9	4.2	4.3	3.5	3.1	3.1	2.9	3.1	3.2	3.4	3.8	(14)
(15) Precipitation	PrecAvg	20.3	1.3	1.1	1.3	1.8	2.2	2.4	2.2	1.3	1.9	1.8	1.3	1.5	(15)
	PrecMax	34.2	4.3	2.6	3.1	5.1	5.7	5.2	6.8	4.4	5.4	3.6	3.2	6.1	(16)
	PrecMin	11.2	0.2	0.1	0.0	0.6	0.7	0.7	0.5	0.2	0.0	0.0	0.0	0.1	(17)
	PrecStd	4.7	0.9	0.7	0.8	0.9	1.1	1.1	1.6	0.9	1.3	1.1	0.9	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	66.3	73.7	83.3	88.0	96.5	97.2	97.9	94.0	83.0	71.8	57.9	(19)
		MCWB	45.5	55.2	56.7	64.0	68.7	74.8	74.7	73.7	71.9	65.9	59.3	49.8	(20)
	2%	DB	48.7	60.5	68.3	77.7	84.3	91.7	93.8	94.4	89.2	76.8	64.2	53.2	(21)
		MCWB	43.3	51.2	54.1	61.6	66.8	72.1	73.2	71.7	67.9	62.2	56.2	47.7	(22)
	5%	DB	44.6	55.4	64.0	72.8	81.3	88.3	90.9	91.5	85.3	72.3	59.0	48.4	(23)
		MCWB	41.0	48.8	51.7	58.8	65.1	70.3	72.2	70.5	66.5	60.4	53.0	44.2	(24)
	10%	DB	40.8	50.9	59.4	69.1	77.9	85.5	88.1	88.8	81.7	67.9	54.5	44.5	(25)
		MCWB	39.0	46.0	49.5	56.6	63.6	69.3	70.9	69.7	65.8	58.4	50.9	41.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.7	56.0	57.9	64.8	70.6	76.1	77.0	76.3	74.6	67.5	59.7	51.4	(27)
		MCDB	52.0	65.8	72.3	80.0	85.7	91.9	92.9	92.9	92.6	77.3	69.5	55.6	(28)
	2%	WB	43.8	52.0	54.7	62.6	68.6	73.5	75.1	74.3	70.9	63.5	56.6	48.0	(29)
		MCDB	47.7	58.6	66.3	76.0	81.7	88.1	90.0	89.9	84.1	73.5	64.3	53.3	(30)
	5%	WB	41.5	49.1	52.6	60.2	66.2	71.9	73.7	72.5	68.5	61.3	54.2	45.0	(31)
		MCDB	44.0	54.6	62.8	71.3	78.2	85.5	87.5	86.8	81.1	70.8	58.6	48.2	(32)
	10%	WB	39.1	46.6	50.2	57.9	64.4	70.5	72.4	71.1	66.5	59.1	51.7	42.1	(33)
		MCDB	40.7	50.5	58.0	66.8	75.6	82.9	84.9	84.6	78.0	66.8	54.7	44.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	16.2	18.9	21.1	21.7	22.2	22.2	23.4	22.5	20.1	12.9	11.9	(35)
		MCDBR	18.2	25.6	29.9	29.6	26.9	27.0	26.9	27.6	29.1	28.4	22.6	19.3	(36)
	5% WB	MCWBR	12.8	16.4	16.6	14.7	11.3	10.1	10.1	9.5	11.4	14.6	13.7	13.4	(37)
		MCWBR	16.1	23.4	27.6	27.8	24.7	25.2	24.0	24.5	26.1	26.2	19.9	18.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.347	0.377	0.415	0.471	0.447	0.453	0.464	0.478	0.443	0.404	0.373	0.351	(40)
		taud	2.285	2.212	2.160	2.018	2.165	2.184	2.160	2.117	2.180	2.275	2.317	2.301	(41)
	Ebn at Noon		240	252	258	252	263	261	256	247	246	240	228	225	(42)
		Edh at Noon	28	35	42	52	46	46	46	47	41	33	27	25	(43)
(44) All-Sky Solar Radiation	RadAvg		462	726	1096	1508	1859	2045	2067	1842	1339	870	493	379	(44)
	RadStd		69	98	109	153	149	134	130	117	104	104	69	53	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (54 neighbors)	+0.85	N/A	N/A	N/A	+0.72	N/A	N/A	-257	+117	N/A

Nomenclature: See separate page